DISCOVERING HIDDEN RELATIONS AMONG TRIANGLE INVARIANTS

TRIANGLE-RELATIONS PROJECT

ABSTRACT. We describe a numerical method for discovering unknown functional relations among scalar quantities derived from a triangle, in the spirit of Euler's relation between the circumradius, the inradius, and the incenter-to- circumcenter distance. A triangle has three degrees of freedom, so a relation among four or more derived scalars is guaranteed by dimension counting alone and is uninformative; a relation among exactly three is a genuine, generically unexpected coincidence. We detect such coincidences by training a bottleneck autoencoder on sampled triangle data and comparing its reconstruction error against a column-permuted null, and make the detector's behavior mathematically precise. Applied to (R, r, d), the method flags exactly this triple as dependent, with no prior knowledge of Euler's relation.

1. DIMENSION COUNTING

A triangle, up to translation and rotation, has three degrees of freedom (e.g. its three side lengths a, b, c); write $\mathcal{M}$ for this moduli space, $\dim \mathcal{M} = 3$. Every derived scalar — area, the circumradius R (the radius of the circle through all three vertices), the inradius r (the radius of the circle inscribed in the triangle), distances between derived points, ... — is a smooth function $f : \mathcal{M} \to \mathbb{R}$, and k of them give a map $F = (f_1, \dots, f_k) : \mathcal{M} \to \mathbb{R}^k$. Call a property of F *generic* if it holds outside a closed, measure-zero exceptional set of choices of the f_i; for instance, the Jacobian $DF(p)$ (the $k \times 3$ matrix of partial derivatives $(\partial f_i / \partial x_j)$ in local coordinates on $\mathcal{M}$ at a point p) generically has the largest rank its shape allows, $\min(k, 3)$, since failing to do so is a closed, codimension-≥ 1 condition on the f_i.

If $k > 3$, take a point p where $DF(p)$ has the generic rank 3, i.e. F is an immersion at p: by the constant rank theorem, in suitable local coordinates near p and $F(p)$, F looks like the standard inclusion $\mathbb{R}^3 \hookrightarrow \mathbb{R}^k$. Consequently the image of a small neighborhood of p is a smooth 3-dimensional submanifold, cut out there by $k - 3$ independent equations $G_i(f_1, \dots, f_k) = 0$ in the $k - 3$ coordinate directions transverse to it. These $k - 3$ local relations arise from dimension counting alone, *for free*, regardless of which f_i were chosen; a relation among four or more triangle invariants therefore says nothing specific about triangle geometry.

If $k = 3$, recall F is a *local diffeomorphism* at p (a smooth bijection near p with smooth inverse) exactly when $DF(p)$ is invertible, by the inverse function theorem. Generically DF has full rank 3, and near such a point (f_1, f_2, f_3) sweeps out an

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open subset of $\mathbb{R}^3$ with nonempty interior — a genuine 3-dimensional volume, not a lower-dimensional surface. An *exact* relation $G(f_1, f_2, f_3) = 0$ holding identically is equivalent to DF having rank ≤ 2 everywhere, collapsing the image onto a 2-dimensional surface: not generic, a genuine algebraic coincidence. Euler's relation

$$d^2 = R^2 - 2Rr \tag{1.1}$$

(where d is the incenter-to-circumcenter distance) is exactly such a statement about (R, r, d). This is the sense in which a relation among exactly three invariants — not four — is structurally new.

Is the solution set of Euler's relation actually a submanifold, or does it have edges or cusps? Generically the former, with one exception.

Proposition 1.1 (Smoothness of the Euler surface). *Let $\Sigma = \{(R, r, d) \in \mathbb{R}^3 : d^2 = R^2 - 2Rr\}$. Then Σ is a smooth (real-analytic) embedded 2-dimensional submanifold of $\mathbb{R}^3$ away from the single point $0 = (0, 0, 0)$, which is a genuine conical singularity: writing $u = R - r$, Euler's relation becomes $u^2 = r^2 + d^2$, the equation of a standard double cone with vertex at $u = r = d = 0$. No actual (nondegenerate) triangle reaches this vertex, since every triangle has $R > 0$ strictly, so the physically realized part of Σ is smooth throughout. It does have one edge — not a cusp — along the equilateral locus $r = R/2$, $d = 0$, where Euler's inequality $R \geq 2r$ becomes an equality.*

Proof. The gradient of $G(R, r, d) = R^2 - 2Rr - d^2$ is $\nabla G = (2R - 2r, -2R, -2d)$, which vanishes only when $R = r$, $R = 0$, $d = 0$ simultaneously, i.e. only at the origin (which does lie on Σ, since $G(0, 0, 0) = 0$). By the implicit function theorem, Σ is a smooth submanifold at every point where $\nabla G \neq 0$, hence everywhere except possibly the origin. Completing the square, $G = (R - r)^2 - r^2 - d^2$, so with $u = R - r$, $\Sigma = \{u^2 = r^2 + d^2\}$: the standard double cone. A punctured neighborhood of its vertex $u = r = d = 0$ on Σ has two connected components (one per nappe, since $u = 0$ forces $r = d = 0$ too), unlike a punctured disk in $\mathbb{R}^2$; so the vertex is not even a topological manifold point, confirming it is a genuine singularity.

For an actual triangle, $R > 0$ always (a nondegenerate triangle has positive circumradius), so the physically realized points of Σ never coincide with the singular vertex. Writing $d = \sqrt{R(R - 2r)}$ (the nonnegative root, since d is a distance) over the physical domain $\{R > 0,\ 0 < r \leq R/2\}$ — using Euler's inequality $R \geq 2r$, which always holds for a genuine triangle — exhibits the physical surface as a smooth graph, with an edge exactly at $r = R/2$ ($d = 0$, the equilateral case). Viewed on the full cone Σ, this locus is an ordinary smooth curve: $\nabla G \neq 0$ there too, e.g. at $(R, R/2, 0)$ with $R > 0$, $\nabla G = (R, -2R, 0) \neq 0$. It becomes an edge of the *physical* surface only because the mirror branch $d < 0$ — equally part of Σ, but physically meaningless — is discarded. □

Figure 1 shows a representative local patch of the physical surface together with sampled triangles, which fall exactly on it, with the equilateral edge marked in red; Figure 2 shows the full variety Σ, both branches, meeting at the singular vertex. (Since Σ is a cone, plotting it from R near 0 up to some $R_{\max}$ in a single view would make it look like a thin flared blade rather than a curved sheet, an artifact of showing many scales at once rather than a feature of the surface itself; Figure 1 instead zooms into one scale.)

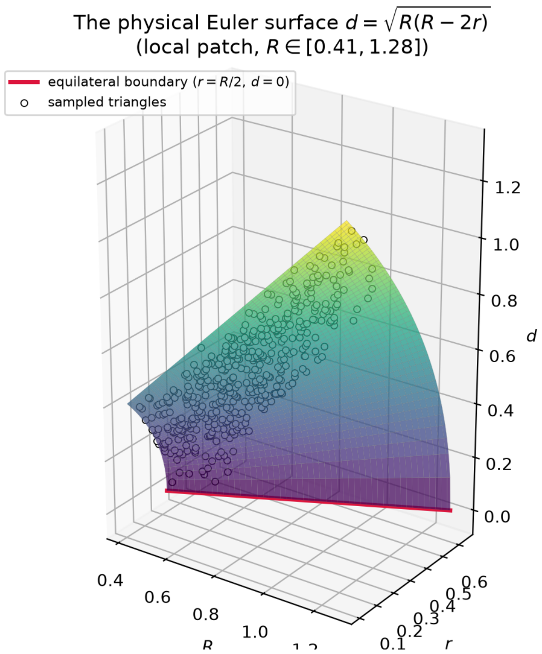

FIGURE 1. The physical Euler surface $d = \sqrt{R(R-2r)}$, shown over a representative local patch of R-values, with sampled triangles (white dots) lying exactly on it. The red line is the equilateral edge $r = R/2$, $d = 0$.

2. The autoencoder detector

Given a candidate triple (f_1, f_2, f_3), sample many random triangles and form the point cloud $X = \{(f_1, f_2, f_3)(T)\} \subset \mathbb{R}^3$. An autoencoder — an encoder $e : \mathbb{R}^3 \to \mathbb{R}^2$ and decoder $d : \mathbb{R}^2 \to \mathbb{R}^3$ trained to minimize $\mathbb{E}\|d(e(x)) - x\|^2$ — searches for the best rank-2 approximation to X: if X lies on a 2-surface, the encoder need only learn a local chart and the decoder its inverse, driving error to 0; if X fills an open 3-region, no map factoring through $\mathbb{R}^2$ can avoid discarding a whole dimension, so error stays bounded away from 0.

2.1. **A precise statement.** Let P be the law of $(f_1, f_2, f_3)(T)$ and $X = \operatorname{supp}(P)$. Fix input and output dimensions $p, q \geq 1$ (for us, always $p, q \in \{2, 3\}$). By a *two-layer perceptron of width h* we mean a map $N : \mathbb{R}^p \to \mathbb{R}^q$ of the form $N(x) = W_2\,\sigma(W_1 x + b_1) + b_2$, with $W_1 \in \mathbb{R}^{h\times p}$, $b_1 \in \mathbb{R}^h$, $W_2 \in \mathbb{R}^{q\times h}$, $b_2 \in \mathbb{R}^q$ trainable and $\sigma = \tanh$ applied componentwise; write $\mathcal{N}_h(\mathbb{R}^p, \mathbb{R}^q)$ for the class of all such maps $\mathbb{R}^p \to \mathbb{R}^q$. Recall g is *K-Lipschitz* if $\|g(x) - g(y)\| \leq K\|x - y\|$; since tanh is 1-Lipschitz, every two-layer perceptron is Lipschitz, with constant controlled by $\|W_1\|, \|W_2\|$. Train an *encoder* $e \in \mathcal{N}_h(\mathbb{R}^3, \mathbb{R}^2)$ and a *decoder* $d \in \mathcal{N}_h(\mathbb{R}^2, \mathbb{R}^3)$ to minimize $L(e, d) = \mathbb{E}_{x\sim P}\|d(e(x)) - x\|^2$.

Proposition 2.1 (Sufficiency)**.** *Recall a subset of $\mathbb{R}^n$ is* compact *iff closed and bounded (Heine–Borel). If X is compact and admits continuous $\varphi : X \to \mathbb{R}^2$, $\psi : \varphi(X) \to \mathbb{R}^3$ with $\psi(\varphi(x)) = x$ for all $x \in X$ — in particular whenever an exact relation confines X to a 2-dimensional set — then for every $\varepsilon > 0$ there is a width h and (e, d) with $L(e, d) < \varepsilon$.*

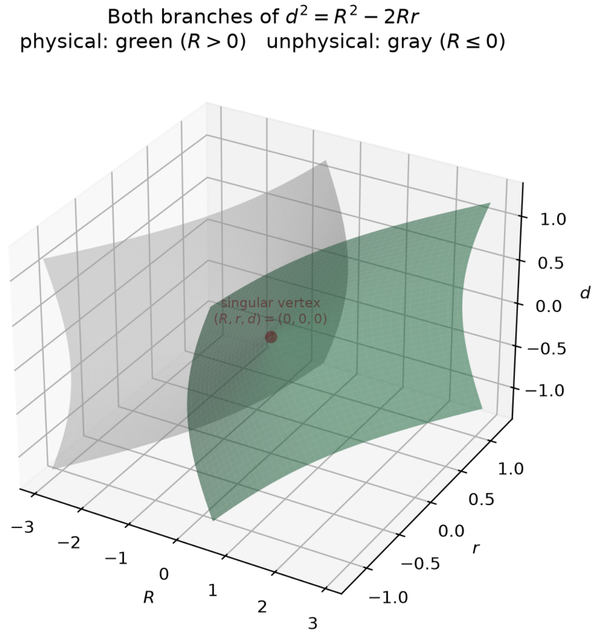

FIGURE 2. Both algebraic branches of $d^2 = R^2 - 2Rr$: the physical branch $R = r + \sqrt{r^2 + d^2}$ (green) and the unphysical branch $R = r - \sqrt{r^2 + d^2} \leq 0$ (gray), meeting only at the singular vertex $(R, r, d) = (0, 0, 0)$.

Proof sketch. By the universal approximation theorem (Cybenko 1989; Hornik 1991): for any continuous $\varphi : K \to \mathbb{R}^q$ on compact K and any $\delta > 0$, some $N \in \mathcal{N}_h(\mathbb{R}^p, \mathbb{R}^q)$ satisfies $\sup_{x \in K} \|N(x) - \varphi(x)\| < \delta$. Approximating φ on X and ψ on $\varphi(X)$ gives e, d with $\sup_{x \in X} \|d(e(x)) - x\| \to 0$, hence $L(e, d) \to 0$. □

In plain terms: if (f_1, f_2, f_3) really do satisfy an exact relation, a large enough autoencoder reconstructs them essentially perfectly, however the relation happens to curve through $\mathbb{R}^3$ — the encoder just learns to read off a valid pair of local coordinates on that curved surface, and the decoder learns to invert them. The next proposition is the converse: without such a relation, no autoencoder of bounded complexity can do better than some fixed, positive error, no matter how it is trained. "Bounded complexity" here means a bound on the Lipschitz constant K of e and d (roughly, how fast their outputs can change relative to their inputs); we explain right after the proof why this qualifier is unavoidable.

Proposition 2.2 (Necessity, at bounded capacity). *Fix $K > 0$. Recall P is* absolutely continuous *(w.r.t. Lebesgue measure) if it assigns zero probability to every Lebesgue-null set, i.e. has a density. If X contains an open ball B on which P is absolutely continuous with $P(B) > 0$, there is $c = c(P|_B, K) > 0$ with $L(e, d) \geq c$ for every pair of K-Lipschitz e, d — in particular for every $(e, d) \in \mathcal{N}_h(\mathbb{R}^3, \mathbb{R}^2) \times \mathcal{N}_h(\mathbb{R}^2, \mathbb{R}^3)$ whose realized weights keep the Lipschitz constant below K.*

Proof sketch. The Hausdorff dimension of $S \subset \mathbb{R}^n$ generalizes ordinary dimension to non-smooth sets (covering S by ε-balls takes $\sim \varepsilon^{-\dim S}$ of them); Lipschitz maps do not increase it. So $d(\mathbb{R}^2)$ has Hausdorff dimension ≤ 2, hence is Lebesgue-null in $\mathbb{R}^3$, hence P-null on B; thus $\mathrm{dist}(x, d(\mathbb{R}^2)) > 0$ for $P|_B$-a.e. x, and a compactness argument over K-Lipschitz maps gives $\mathbb{E}_{x\sim P|_B} \mathrm{dist}(x, d(\mathbb{R}^2))^2 \geq c > 0$. Since $d(e(\mathbb{R}^3)) \subseteq d(\mathbb{R}^2)$, the same bound applies to $L(e, d)$. $\square$

Remark 2.3. By Brouwer's invariance-of-domain theorem, a continuous injection between open subsets of $\mathbb{R}^n$ is an open map, so no continuous injection of $\mathbb{R}^2$ into $\mathbb{R}^3$ has open image: exact space-filling is topologically impossible regardless of capacity. What is not ruled out is *approximate* space-filling by increasingly large-Lipschitz-constant maps as $K \to \infty$, which is why Proposition 2.2 fixes K; in practice ℓ_2 weight regularization and early stopping keep the realized Lipschitz constant controlled.

Propositions 2.1 and 2.2 say that, at any fixed, regularized capacity, reconstruction error going to 0 versus staying bounded away from it is decided entirely by whether (f_1, f_2, f_3) satisfy an exact relation — the theoretical content behind the detection rule used throughout. Figure 3 shows this dichotomy on the actual triple (R, r, d).

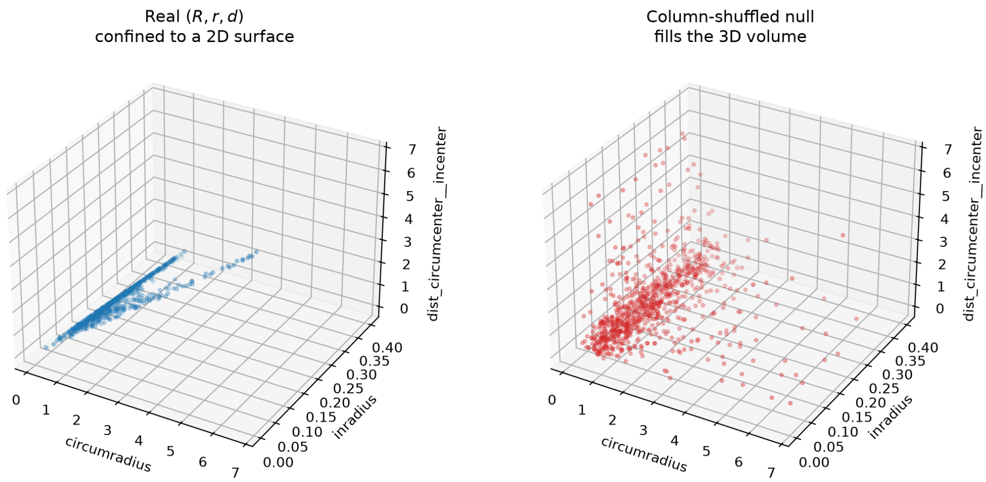

FIGURE 3. The real triple (R, r, d) (left, 1000 sampled triangles) collapses onto a thin 2-dimensional surface, as Proposition 2.1 predicts; independently shuffling each column (right) destroys the relation, and the same points fill an open 3-dimensional region, as Proposition 2.2 predicts.

3. Calibration via a permutation null

Propositions 2.1 and 2.2 describe an idealized limit ($h \to \infty$, exact expectations); in practice we train one finite network on one finite sample, and its reconstruction loss L_{real} is never exactly 0 or exactly the theoretical floor c — it is a single noisy number. An absolute threshold on L_{real} ("flag it as dependent if the loss is below 0.01", say) is therefore not meaningful: different triples of scalars differ in units and scale, network training is itself noisy, and, since all derived scalars are functions of the same three underlying triangle parameters, even functionally *independent* triples can be mildly correlated and so reconstruct a bit better than one might naively expect. What we need is not one global cutoff but a baseline for "no relation" calibrated separately for each triple.

We obtain that baseline by destroying exactly the joint structure between the three scalars while leaving everything else about the data unchanged. Concretely, write the sample as a matrix $X \in \mathbb{R}^{n\times 3}$, one row per sampled triangle. Form a *shuffled* copy X' by independently and uniformly permuting the n entries within each of the three columns (a different random permutation per column). Each column of X' is some reordering of the corresponding column of X, so it has exactly the same values, and hence the same one-dimensional marginal distribution, as before; but the pairing *across* columns is now uniformly random, so whatever functional relationship held between f_1, f_2, f_3 in the real data has been destroyed. Training the same encoder/decoder architecture on X' therefore measures how well an autoencoder can compress three quantities that merely share these marginal distributions but have no functional relationship at all — precisely the situation of Proposition 2.2.

We repeat the shuffle-and-train step m times (an independent random reshuffling of X each time), obtaining m null reconstruction losses $L'_1, \ldots, L'_m$, and record their sample mean and sample standard deviation,

$$\mu_{\text{null}} = \frac{1}{m}\sum_{i=1}^{m} L'_i, \qquad \sigma_{\text{null}} = \left(\frac{1}{m-1}\sum_{i=1}^{m}(L'_i - \mu_{\text{null}})^2\right)^{1/2}. \tag{3.1}$$

These summarize the null distribution of reconstruction loss for this particular triple's marginals: μ_{null} is its typical value under "no relation", and σ_{null} is how much that value fluctuates from one reshuffling (and training run) to the next. We compare the real loss to this null distribution via the standard z-score,

$$z = \frac{\mu_{\text{null}} - L_{\text{real}}}{\sigma_{\text{null}}}, \tag{3.2}$$

i.e. the number of null standard deviations by which the real loss falls below the null mean, or equivalently via the ratio $L_{\text{real}}/\mu_{\text{null}}$. A large z (ratio close to 0) means the real triple reconstructs far better than shuffled versions of itself typically do: the fingerprint of Proposition 2.1 taking over, and our evidence of an unexpected functional relation. A z near 0 (ratio close to 1) means the real data compresses no better than its own null, consistent with Proposition 2.2 and no relation. Because μ_{null} and σ_{null} are computed separately for each candidate triple, from data with the same units, scale, and marginal shape as the real sample, this comparison is self-calibrating and needs no triple-independent threshold.

Figure 4 shows this construction on (R, r, d) itself, with $m = 20$ reshufflings: L_{real} sits well to the left of every one of the m null losses, and in particular below the $\mu_{\text{null}} - \sigma_{\text{null}}$ edge of the shaded band.

4. The pipeline

(1) Enumerate candidate triples of derived scalars.
(2) Sample random triangles once per triple; train a 2-bottleneck autoencoder; compare held-out error to a column-shuffled null (several repeats).
(3) Rank triples by real/null ratio (small = strong candidate).

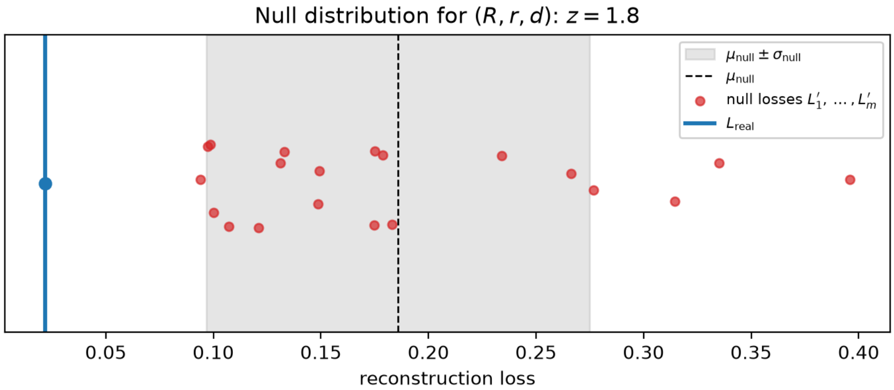

FIGURE 4. The null distribution of reconstruction loss for (R, r, d): $m = 20$ column-shuffled losses $L'_1, \ldots, L'_m$ (red), their mean μ_{null} and ± 1 standard-deviation band (dashed line, shaded), and the real loss L_{real} (blue), which lands clearly below the entire null cluster.

The autoencoder is purely a screening tool: it flags *whether* a triple is suspiciously dependent, not what the relation is. The `verify_euler_relation` module runs this on (R, r, d), confirming it flags Euler's relation with no prior knowledge of the formula.